

number		
8(a)(i)	C galactose and glucose A is incorrect because lactose consists of glucose and galactose monomers B is incorrect because lactose consists of glucose and galactose monomers Dis incorrect because lactose consists of glucose and galactose monomers	(1)

Question number	Answer	Additional guidance	Mark
8(a)(ii)	An explanation that includes three of the following points: - lactase is soluble because of its {globular shape / external polar R groups} (1) (and therefore) lactase collides with lactose (1) - active site of lactase is complementary to the lactose (1) - Formation of enzyme-substrate complex lowers the activation energy (1)	Accept hydrophilic for polar Accept active site of lactase binds to lactose / active site allows enzyme-substrate complex to form Accept R group interactions break the glycosidic bonds	(3)

Question number	Answer	Additional guidance	Mark
8(b)(i)	{lactase / enzyme} is reusable / milk is not contaminated with {lactase / enzyme} (1)	Accept higher enzyme activity	(1)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	An explanation that includes four of the following points: - pH {below 5 / above 5} reduces lactase activity (1) - because pH affects the shape of the active site (1) - due to ionisation of the R groups (1) - immobilised lactase is active at wider range of pH values (1) - immobilisation holds the R groups in place so active site does not change shape (1)	Accept pH 5 is the optimum for both enzymes / pH 4-6 is the optimum for both enzymes Accept active site denatured by pH Accept bonds between R groups are broken	(4)

number																		
8(b)(iii)	An answer that includes the following points: • measure the decrease in concentration of lactose over time • see table for units OR • measure the increase in concentration of {glucose / galactose} over time • see table for units	Accept measure rate of loss of lactose Ignore: measure how long it takes for lactose to be fully broken down Accept measure rate of production of {glucose / galactose} Units can be expressed as: • mass per volume per time • mass volume⁻¹ time⁻¹ • amount per volume per time • amount volume⁻¹ time⁻¹ <table><tr><td>mass</td><td>volume</td><td>time</td></tr><tr><td>g / mg / µg / kg</td><td>cm⁻³ / mm⁻³ / dm⁻³ / litre⁻¹</td><td>s⁻¹ / min⁻¹ / hour⁻¹</td></tr><tr><td></td><td></td><td></td></tr><tr><td>amount</td><td>volume</td><td>time</td></tr><tr><td>mmol / mol / µmol</td><td>cm⁻³ / mm⁻³ / dm⁻³ / litre⁻¹</td><td>s⁻¹ / min⁻¹ / hour⁻¹</td></tr></table>	mass	volume	time	g / mg / µg / kg	cm ⁻³ / mm ⁻³ / dm ⁻³ / litre ⁻¹	s ⁻¹ / min ⁻¹ / hour ⁻¹				amount	volume	time	mmol / mol / µmol	cm ⁻³ / mm ⁻³ / dm ⁻³ / litre ⁻¹	s ⁻¹ / min ⁻¹ / hour ⁻¹	(2)
mass	volume	time																
g / mg / µg / kg	cm ⁻³ / mm ⁻³ / dm ⁻³ / litre ⁻¹	s ⁻¹ / min ⁻¹ / hour ⁻¹																
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mmol / mol / µmol	cm ⁻³ / mm ⁻³ / dm ⁻³ / litre ⁻¹	s ⁻¹ / min ⁻¹ / hour ⁻¹																

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8(c)	An answer that includes two of the following points: • mutation (that resulted in CLI) occurred in the {gene / DNA} (of people living in one country) (1) • people from this country (had children that) stayed in this country (1) • relatively new mutation so has not had the chance to spread (1)	Accept limited emigration from this country / reproduction with others from same country Ignore idea that mutation is caused by drinking milk / not drinking milk	(2)